

Sprawozdanie

Charakterystyki regulatorów

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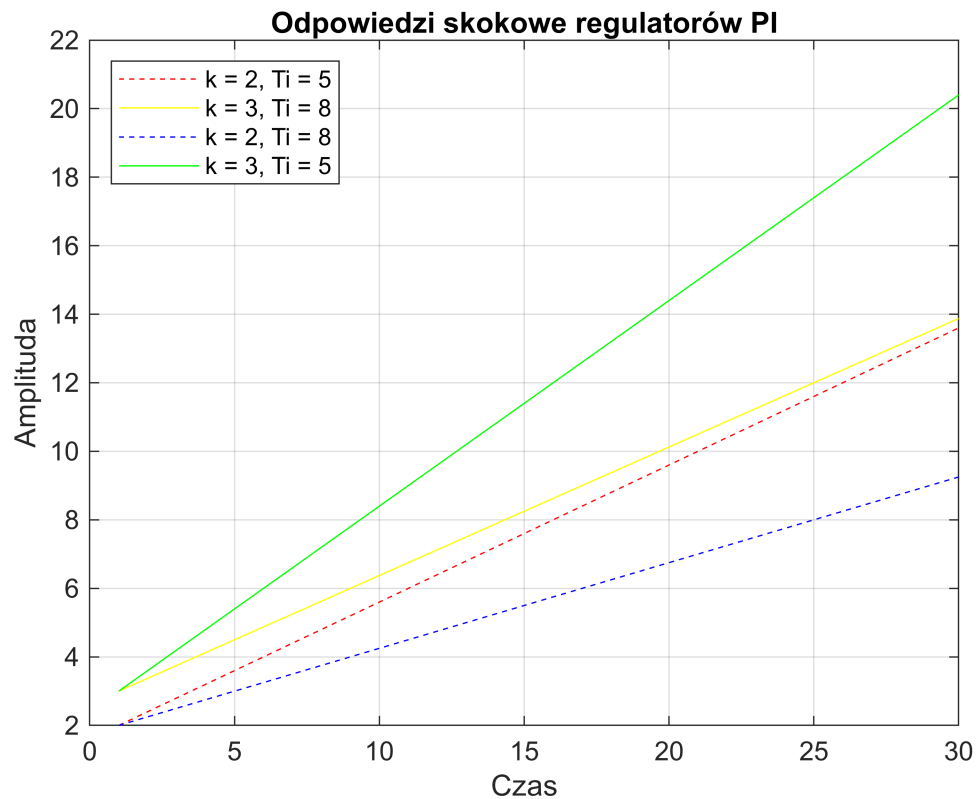
Cel ćwiczenia:

Celem ćwiczenia było sporządzenie charakterystyk regulatorów czasowych i częstotliwościowych (skokowe, nyquist, bode), dla regulatorów PI, PD, PID.

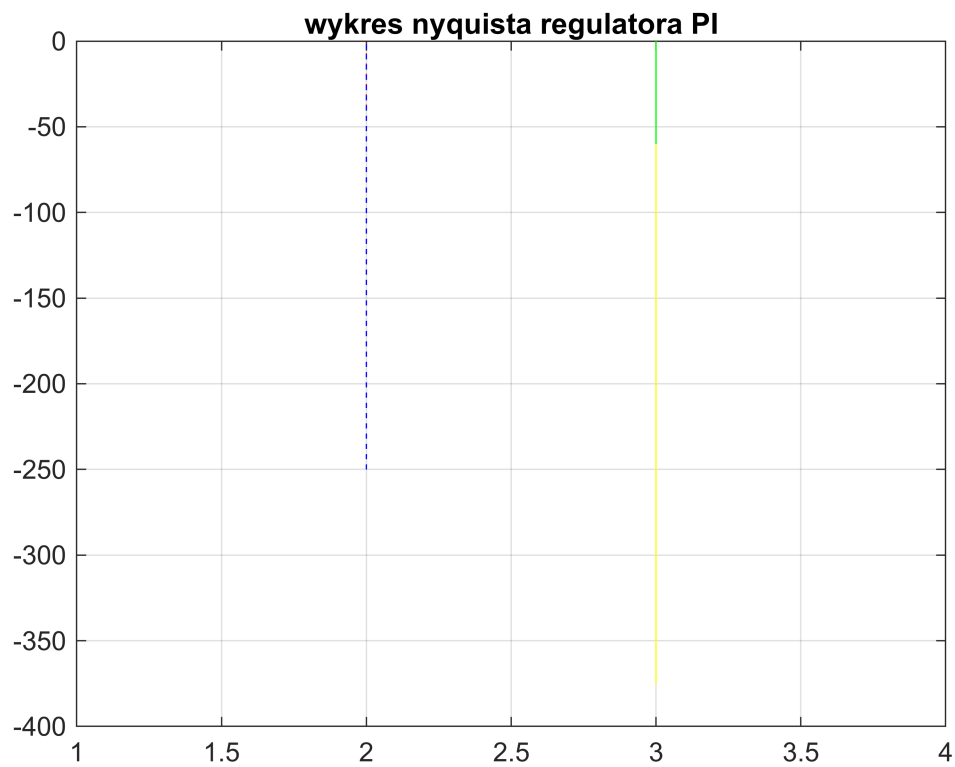
Regulator PI

```
% KTis + K/Tis
%transmitancje
%1) k = 2 Ti = 5
t1 = linspace(1,30);
K = 2;
Ti = 5;
licznik1_1 = [K*Ti,K];
mianownik1_1 = [Ti ,0];
sys1_1 = tf(licznik1_1,mianownik1_1);
%2) K = 3 Ti = 8
K = 3;
Ti = 8;
licznik1_2 = [K*Ti,K];
mianownik1_2 = [Ti ,0];
sys1_2 = tf(licznik1_2,mianownik1_2);
%3) K = 2 Ti = 8
K = 2;
Ti = 8;
licznik1_3 = [K*Ti,K];
mianownik1_3 = [Ti ,0];
sys1_3 = tf(licznik1_3,mianownik1_3);
%4) K = 3 Ti = 5
K = 3;
Ti = 5;
licznik1_4 = [K*Ti,K];
mianownik1_4 = [Ti ,0];
sys1_4 = tf(licznik1_4,mianownik1_4);
%charakterystyki skokowe
step1_1 = step(sys1_1,t1);
step1_2 = step(sys1_2,t1);
step1_3 = step(sys1_3,t1);
step1_4 = step(sys1_4,t1);
plot(t1,step1_1,'r--',t1,step1_2,'y-',t1,step1_3,...
     , 'b--',t1,step1_4,'g-')
legend('k = 2, Ti = 5', 'k = 3, Ti = 8', 'k = 2, Ti = 8', 'k = 3, Ti = 5',
Location='northwest');
grid on;
title('Odpowiedzi skokowe regulatorów PI');
```

```
xlabel('Czas');
ylabel('Amplituda');
```

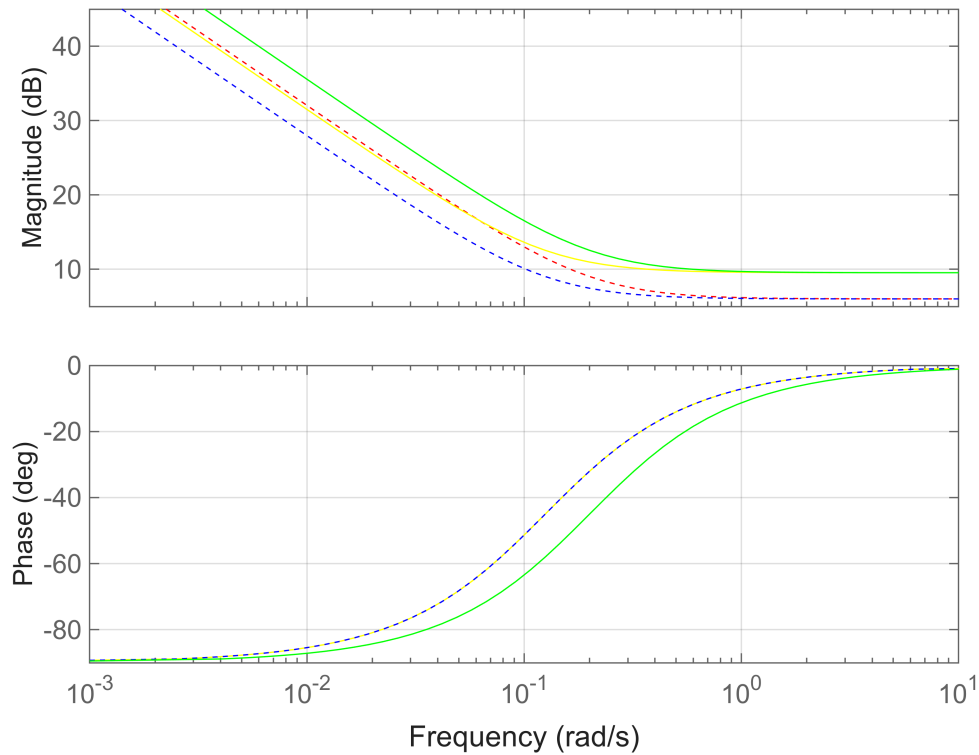


```
%nyquist
[re1_1,im1_1] = nyquist(sys1_1);
[re1_2,im1_2] = nyquist(sys1_2);
[re1_3,im1_3] = nyquist(sys1_3);
[re1_4,im1_4] = nyquist(sys1_4);
plot(re1_1(:),im1_1(:),'r--',re1_2(:),im1_2(:),'y-',re1_3(:),im1_3(:),'b--',re1_4(:),im1_4(:),'g-')
axis([1, 4, -400, 0]);
title('wykres nyquista regulatora PI');
grid on;
```



```
%bode
[mag1_1,phase1_1] =bode(sys1_1);
[mag1_2,phase1_2] =bode(sys1_2);
[mag1_3,phase1_3] =bode(sys1_3);
[mag1_4,phase1_4] =bode(sys1_4);
bodeplot(sys1_1,'r--');
hold on;
bodeplot(sys1_2,'y-');
hold on;
bodeplot(sys1_3,'b--');
hold on;
bodeplot(sys1_4,'g-');
hold off;
title('wykres bodego regulatora PI');
grid on;
```

wykres bodego regulatora PI



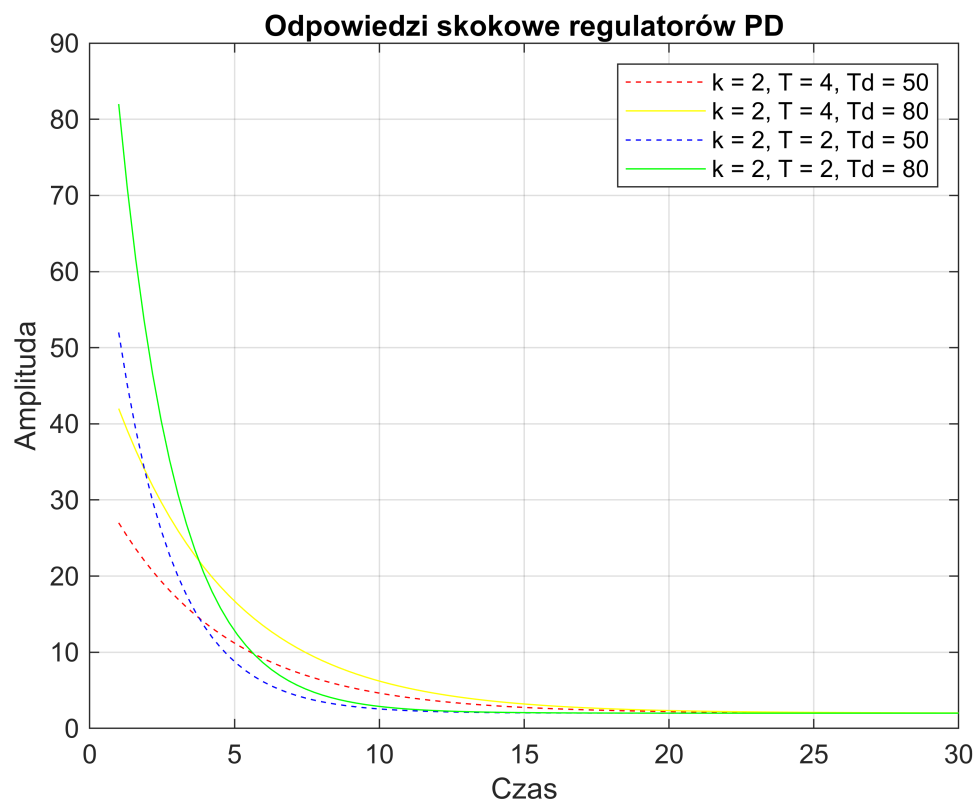
Regulator PD

```
% K((Ts+Tds)+1)/Ts +1 , T < 0.1Td
%1) K = 2, T =4, Td =50
K = 2;
T = 4;
Td = 50;
licznik2_1 = [K*T + K*Td, K];
mianownik2_1 = [T,1];
sys2_1 = tf(licznik2_1,mianownik2_1);
%2) K =2, T = 4, Td= 80
K = 2;
T =4;
Td = 80;
licznik2_2 = [K*T + K*Td, K];
mianownik2_2 = [T,1];
sys2_2 = tf(licznik2_2,mianownik2_2);
%3) K=2, T =2, Td = 50
K=2;
T=2;
Td=50;
licznik2_3 = [K*T + K*Td, K];
mianownik2_3 = [T,1];
sys2_3 = tf(licznik2_3,mianownik2_3);
%4) K=2, T=2, Td=80
```

```

K=2;
T=2;
Td =80;
licznik2_4 = [K*T + K*Td, K];
mianownik2_4 = [T,1];
sys2_4 = tf(licznik2_4,mianownik2_4);
%charakterystyki skokowe
step2_1 = step(sys2_1,t1);
step2_2 = step(sys2_2,t1);
step2_3 = step(sys2_3,t1);
step2_4 = step(sys2_4,t1);
plot(t1,step2_1,'r--',t1,step2_2,'y-',t1,step2_3 ...
    ,'b--',t1,step2_4,'g-')
legend('k = 2, T = 4, Td = 50', 'k = 2, T = 4, Td = 80', 'k = 2, T = 2, Td = 50',
    'k = 2, T = 2, Td = 80');
grid on;
title('Odpowiedzi skokowe regulatorów PD');
xlabel('Czas');
ylabel('Amplituda');

```

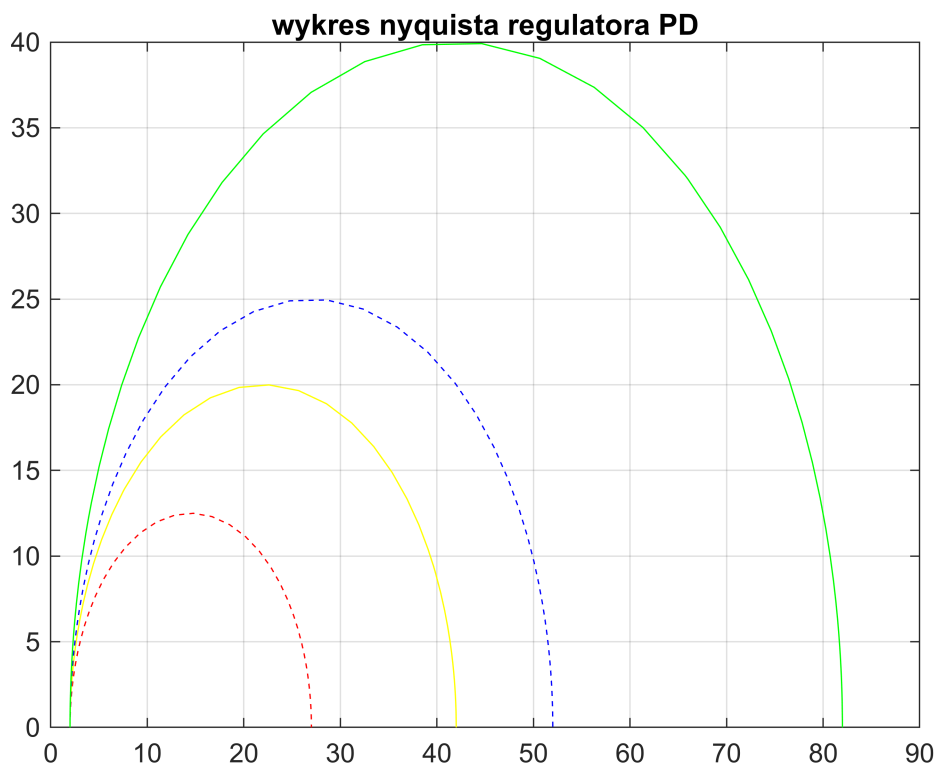


```

%nyquist
[re2_1, im2_1] = nyquist(sys2_1);
[re2_2, im2_2] = nyquist(sys2_2);
[re2_3, im2_3] = nyquist(sys2_3);
[re2_4, im2_4] = nyquist(sys2_4);
plot(re2_1(:),im2_1(:),'r--',re2_2(:),im2_2(:),'y-',re2_3(:),im2_3(:),'b--',re2_4(:),
    im2_4(:),'g-')

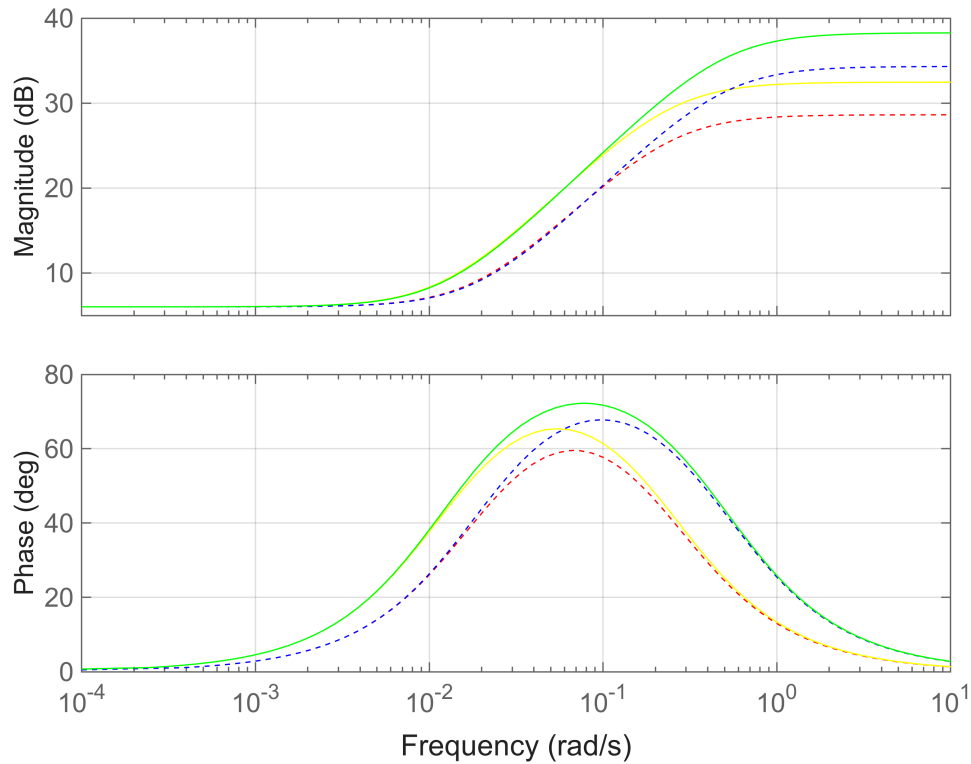
```

```
title('wykres nyquista regulatora PD');
grid on;
```



```
%bode
bodeplot(sys2_1,'r--');
hold on;
bodeplot(sys2_2,'y-');
hold on;
bodeplot(sys2_3,'b--');
hold on;
bodeplot(sys2_4,'g-');
hold off;
title('wykres bodego regulatora PD');
grid on;
```

wykres bodego regulatora PD



Regulator PID

%1) $K=4, T_i=5, T=4, T_d=50$

$K=4, T_i=5, T=4, T_d=50$;

```
K =
4
Ti =
5
T =
4
```

```
licznik3_1 = [K * Ti * T + K * Ti * Td, K * Ti + K * T, K];
```

```
mianownik3_1 = [Ti*T,Ti,0];
```

```
sys3_1 = tf(licznik3_1,mianownik3_1);
```

%2) $K=4, T_i=6, T=4, T_d=80$

$K=4, T_i=6, T=4, T_d=80$;

```
K =
4
Ti =
6
T =
4
```

```
licznik3_2 = [K * Ti * T + K * Ti * Td, K * Ti + K * T, K];
```

```
mianownik3_2 = [Ti*T,Ti,0];
```

```
sys3_2 = tf(licznik3_2,mianownik3_2);
```

%3) $K=4, T_i=7, T=2, T_d=50$

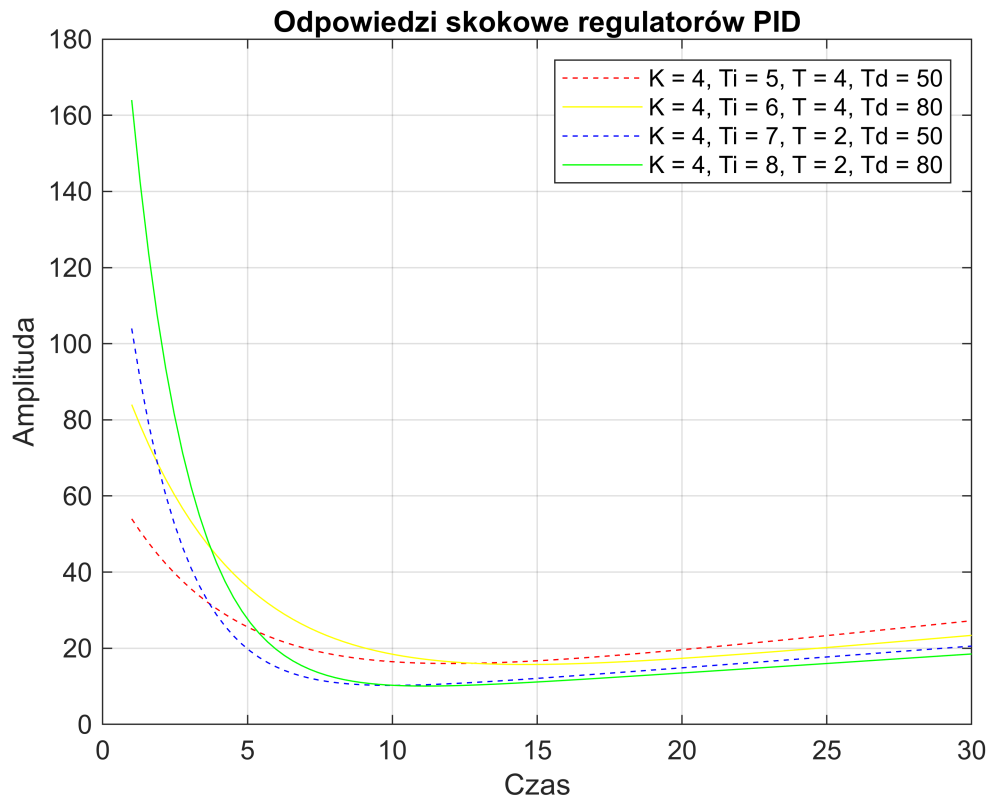
```
K=4,Ti=7,T=2,Td=50;
```

```
K =  
4  
Ti =  
7  
T =  
2
```

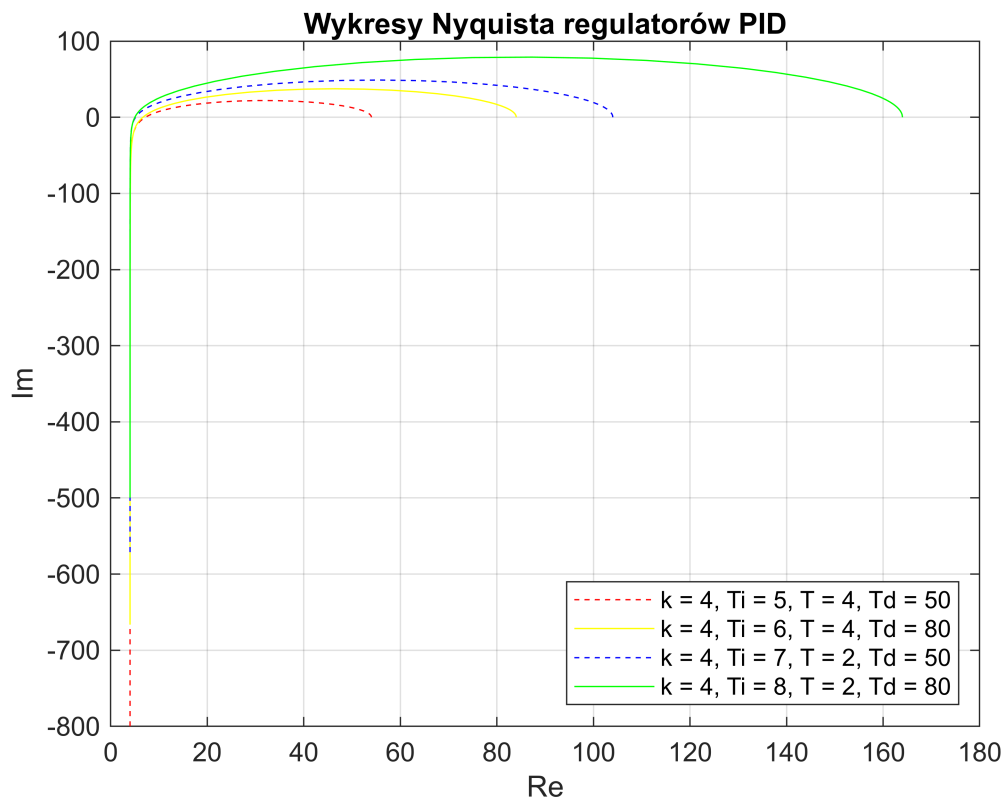
```
licznik3_3 = [K * Ti * T + K * Ti * Td, K * Ti + K * T, K];  
mianownik3_3 = [Ti*T,Ti,0];  
sys3_3 = tf(licznik3_3,mianownik3_3);  
%1) K=4,Ti=8,T=2,Td=80  
K=4,Ti=8,T=2,Td=80;
```

```
K =  
4  
Ti =  
8  
T =  
2
```

```
licznik3_4 = [K * Ti * T + K * Ti * Td, K * Ti + K * T, K];  
mianownik3_4 = [Ti*T,Ti,0];  
sys3_4 = tf(licznik3_4,mianownik3_4);  
%charakterystyki skokowe  
step3_1 = step(sys3_1,t1);  
step3_2 = step(sys3_2,t1);  
step3_3 = step(sys3_3,t1);  
step3_4 = step(sys3_4,t1);  
plot(t1,step3_1,'r--',t1,step3_2,'y-',t1,step3_3 ...  
    , 'b--',t1,step3_4,'g-')  
legend('K = 4, Ti = 5, T = 4, Td = 50', 'K = 4, Ti = 6, T = 4, Td = 80', 'K = 4, Ti  
= 7, T = 2, Td = 50', 'K = 4, Ti = 8, T = 2, Td = 80');  
grid on;  
title('Odpowiedzi skokowe regulatorów PID');  
xlabel('Czas');  
ylabel('Amplituda');
```

```
%nyquist
[re3_1, im3_1] = nyquist(sys3_1);
[re3_2, im3_2] = nyquist(sys3_2);
[re3_3, im3_3] = nyquist(sys3_3);
[re3_4, im3_4] = nyquist(sys3_4);
plot(re3_1(:),im3_1(:),'r--',re3_2(:),im3_2(:),'y-',re3_3(:),im3_3(:),'b--',re3_4(:)
,im3_4(:),'g-')
legend('k = 4, Ti = 5, T = 4, Td = 50', 'k = 4, Ti = 6, T = 4, Td = 80', 'k = 4, Ti
= 7, T = 2, Td = 50', 'k = 4, Ti = 8, T = 2, Td = 80', Location='southeast');
grid on;
title('Wykresy Nyquista regulatorów PID');
xlabel('Re');
ylabel('Im');
```



```
%bode
bodeplot(sys3_1, 'r--');
hold on;
bodeplot(sys3_2, 'y-');
hold on;
bodeplot(sys3_3, 'b--');
hold on;
bodeplot(sys3_4, 'g-');
hold off;
title('wykres bodego regulatora PID');
grid on;
```

wykres bodego regulatora PID

